

10.5.2

$$X_{n+1} = 10 X_n \pmod{1}$$

$$\lambda = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=0}^{n-1} \ln |f'(x_i)|$$

$$f'(x_i) = 10, \text{ so}$$

$$\lambda = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=0}^{n-1} \ln |10|$$

$$\lambda = \ln[10]$$

10.5.7

Suppose x^* is a fixed point of f undergoing a period-doubling bifurcation, then $f(x^*) = -1$. Regarding x^* as x_0 , we see $x_1 = x_2 = x_3 = \dots = x^*$ since x^* is fixed. Hence,

$$\begin{aligned}\lambda &= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=0}^{n-1} \ln |f'(x_i)| = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=0}^{n-1} \ln |f'(x^*)| \quad n \text{ terms} \\ &= \lim_{n \rightarrow \infty} \frac{1}{n} n \ln |f'(x^*)| = \lim_{n \rightarrow \infty} \ln |f'(x^*)| = \ln |f'(x^*)| = \ln |-1| \\ &= \ln(1) = 0\end{aligned}$$

This shows $\lambda = 0$ at $r = r_1$ for the logistic map.

Next, consider where period-2 bifurcates to period-4: At $r = r_2$, we have period-2 points p, q with $f(p) = q$, $f(q) = p$ and $(f^2)'(p) = -1$, $(f^2)'(q) = -1$, so $f'(f(p))f'(q) = -1$, i.e. $f'(p)f'(q) = -1$.

Consider the sequence $\{a_n\}$ in the definition of Liapunov exponent:

$$\begin{aligned}\lim_{n \rightarrow \infty} a_n &= \lambda \quad a_n = \frac{1}{n} \ln \left(\prod_{i=0}^{n-1} |f'(x_i)| \right) \\ a_1 &= \frac{1}{1} \ln (|f'(x_0)|) = \ln (|f'(p)|) \\ a_2 &= \frac{1}{2} \ln (|f'(x_0)||f'(x_1)|) = \frac{1}{2} \ln (|f'(p)f'(q)|) \\ &= \frac{1}{2} \ln (|-1|) = 0 \quad \text{Since } f'(p)f'(q) = -1 \text{ when } r = r_2 \\ a_3 &= \frac{1}{3} \ln (|f'(x_0)f'(x_1)f'(x_2)|) \\ &= \frac{1}{3} \ln (|f'(p)f'(q)f'(p)|) = \frac{1}{3} \ln (|(-1)f'(p)|) \\ &= \frac{1}{3} \ln |f'(p)|\end{aligned}$$

In general

$$\begin{cases} a_{2k+1} = \frac{1}{2k+1} \ln |f'(p)| \\ a_{2k} = 0 \end{cases}$$

and as $k \rightarrow \infty$, this means $a_{2k+1} \rightarrow 0$ and therefore $a_n \rightarrow 0$ as $n \rightarrow \infty \Rightarrow \lambda = 0$ at $r = r_2$.

The same idea works at higher r_m .

11.3.3

a)

One iteration of the even-sevenths Cantor set leaves four intervals that will each contain the even-sevenths Cantor set scaled by a factor of seven. Then the similarity dimension is

$$4 = 7^d \Rightarrow d = \frac{\ln(4)}{\ln(7)}$$

b)

For n odd, one iteration of the even- n 'ths Cantor set leaves $\frac{n+1}{2}$ intervals that will each contain the even- n 'ths Cantor set scaled by a factor of n . Then the similarity dimension is

$$\frac{n+1}{2} = n^d \Rightarrow d = \frac{\ln(n+1) - \ln(2)}{\ln(n)}$$

11.3.5

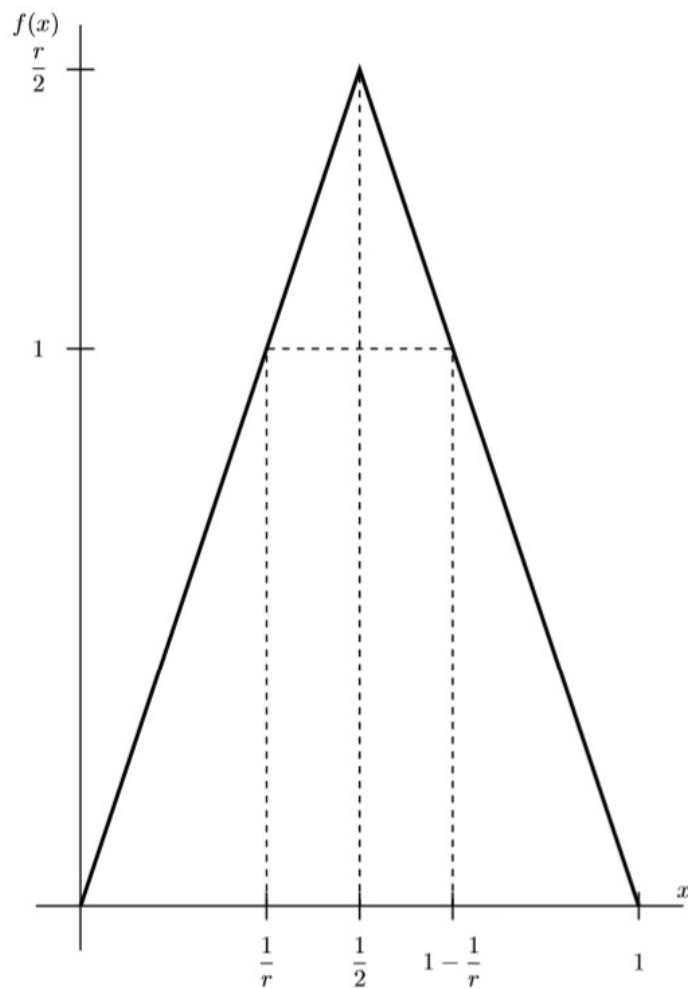
One iteration of this no 8's Cantor set will remove the interval $[0.8, 0.9)$ from $[0, 1]$ and leave nine intervals that will each contain this no 8's Cantor set scaled by a factor of ten. Then the similarity dimension is

$$9 = 10^d \Rightarrow d = \frac{\ln(9)}{\ln(10)}$$

11.4.6

a)

Making a graph is quite useful.

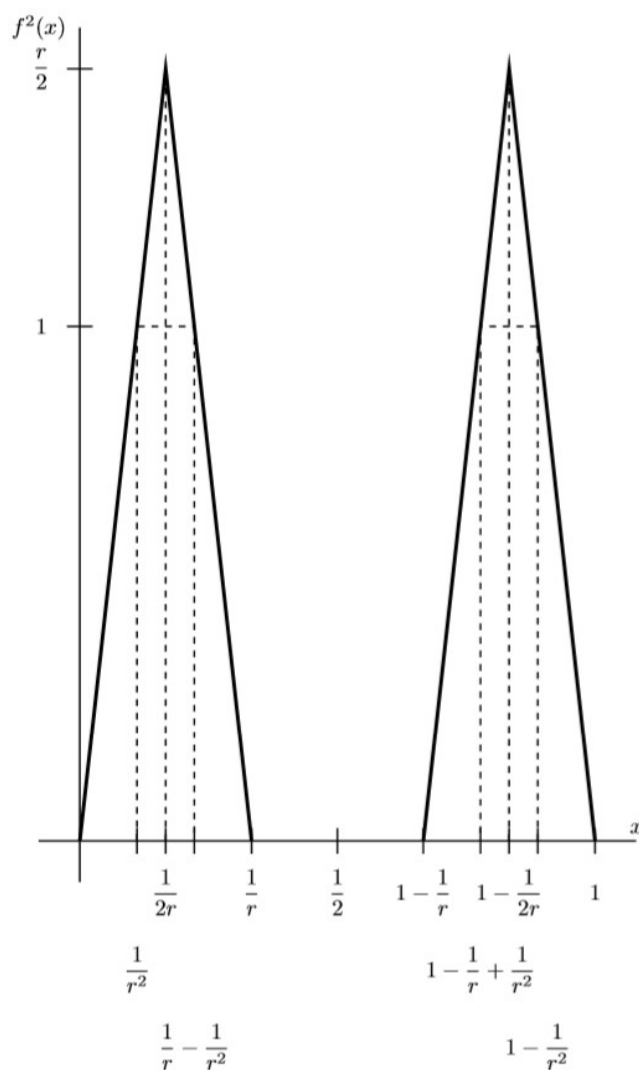


From the graph we can already see the intervals that escape after one iteration. Doing some algebra to check,

$$\begin{aligned} 0 \leq x_0 \leq \frac{1}{2} \quad f(x_0) = rx_0 > 1 &\Rightarrow \frac{1}{r} < x_0 \\ \frac{1}{2} \leq x_0 \leq 1 \quad f(x_0) = r(1 - x_0) > 1 &\Rightarrow x_0 < 1 - \frac{1}{r} \\ x_0 \in \left(\frac{1}{r}, 1 - \frac{1}{r}\right) &\Rightarrow f(x_0) > 1 \end{aligned}$$

The intervals that escapes after two iterations is not as easy to see from the first graph, but from the iterated graph they are easier to see.

11.4.6 (cont.)



(The middle portion has been left out since those points escape by the first iteration.)

And if we do the algebra,

$$\begin{aligned} f^2(x_0) = x_2 > 1 \text{ and } 0 \leq f(x_0) = x_1 \leq 1 \\ \Rightarrow f(x_0) \in \left(\frac{1}{r}, 1 - \frac{1}{r}\right) \end{aligned}$$

since x_1 has to land in $[0, 1]$ and then x_2 has to escape.

$$0 \leq x_0 \leq \frac{1}{2} \quad \frac{1}{r} < f(x_0) = rx_0 < 1 - \frac{1}{r} \Rightarrow \frac{1}{r^2} < x_0 < \frac{1}{r} - \frac{1}{r^2}$$

11.4.6 (Cont.)

$$\begin{aligned} \frac{1}{2} \leq x_0 \leq 1 \quad \frac{1}{r} < f(x_0) = rx_0 < 1 - \frac{1}{r} &\Rightarrow 1 - \frac{1}{r} + \frac{1}{r^2} < x_0 < 1 - \frac{1}{r^2} \\ x_0 \in \left(\frac{1}{r^2}, \frac{1}{r} - \frac{1}{r^2} \right) \cup \left(1 - \frac{1}{r} + \frac{1}{r^2}, 1 - \frac{1}{r^2} \right) \\ \Rightarrow f(x_0) \in \left(\frac{1}{r}, 1 - \frac{1}{r} \right) \\ \Rightarrow f^2(x_0) = f(x_1) > 1 \end{aligned}$$

b)

Trying to find the set of points that never escape purely by algebra looks like a royal mess. Visible in the graph though is the recursive nature of the iterated function. The problem repeats in small intervals, where points escape from the middle of each interval. We're creating the middle- $(1 - \frac{2}{r})$ 'ths Cantor set. You can show this analytically by horizontally scaling and shifting in each interval of the iterated graph to transform the domain into $[0,1]$ which gives the $f(x)$ graph. Hence the recursive procedure is exactly the same as the middle- $(1 - \frac{2}{r})$ 'ths Cantor set.

Therefore, points in the middle- $(1 - \frac{2}{r})$ 'ths Cantor set never escape within a finite number of iterations.

c)

For every iteration of constructing the middle- $(1 - \frac{2}{r})$ 'ths Cantor set, each interval to be descended into is divided into two new intervals with the length reduced by a factor of r . From this we can calculate

$$\epsilon = \left(\frac{1}{r} \right)^n \quad N(\epsilon) = 2^n$$

and the box dimension is

$$d = \lim_{\epsilon \rightarrow 0} \frac{\ln(N(\epsilon))}{\ln\left(\frac{1}{\epsilon}\right)} = \lim_{n \rightarrow \infty} \frac{\ln(2^n)}{\ln(r^n)} = \frac{\ln(2)}{\ln(r)}$$

d)

The Liapunov exponent is defined as

$$\lambda = \frac{1}{n} \lim_{n \rightarrow \infty} \sum_{i=0}^{n-1} \ln |f'(x_i)|$$

and since we're in the invariant set, x_i is always in the domain of $f(x)$ and the magnitude of the slope is constant.

$$\lambda = \frac{1}{n} \lim_{n \rightarrow \infty} \sum_{i=0}^{n-1} \ln(r) = \ln(r) \quad r > 2 \Rightarrow \ln(r) > 0$$

Therefore the Liapunov exponent is positive at each point in the invariant set.

12, 2 .18

$$a = 1.7 \quad b = 0.5$$

